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An Individual Risk Model for Premium Calculation Based on Quantile: A Comparison between Generalized Linear Models and Quantile Regression

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This article deals with the use of quantile regression and generalized linear models for a premium calculation based on quantiles. A premium principle is a functional that assigns a usually loaded premium to any distribution of claims. The loaded premium is generally greater than the expected value of the loss and the difference is considered to be a risk margin or a safety loading. The failure of a right charge of individual risk rate exposes the insurer to adverse selection and, consequently, to deteriorating financial results. The article aim is to define the individual pure premium rates and the corresponding risk margin in balance with some profit or solvency constraints. We develop a ratemaking process based on a two-part model using quantile regression and a gamma generalized linear model, respectively, in order to estimate the claim's severity quantiles. Generalized linear models focus on the estimation of the mean of the conditional loss distribution but have some drawbacks in assessing distribution moments other than the mean and are very sensitive to outliers. On the contrary, quantile regression exceeds these limits leading to the estimate of conditional quantiles and is more accurate for a better measurement of the variability of a risk class. The proposed methodology for premium calculation allows us to find further limits of the generalized linear model as the solution we found, under specific assumptions for a generalized linear model, is equivalent to the application of the expected value premium principle. Finally, the methodology we suggest for the two-part quantile regression allows reducing the practical issue of overmodeling and over-parameterization with respect to other proposals on the same topic.

1. INTRODUCTION

The assessment of underwriting risk is a key task in the risk management framework and is central in order to value the adequacy of premiums to cover costs and expenses associated with new or unexpired insured risks. It originates from the uncertainty related to the number and size of future claims to be settled compared to expected ones. In its valuation it is a standard practice to consider a risk margin (or safety loading) for adverse deviations compared to the expected value. The need for an implicit or explicit safety loading is a basic concept in the actuarial valuations and it is justified from a theoretical point of view by risk theory (see Olivieri and Pitacco 2011). Moreover, the existence of a risk margin is well known in practice and it is considered crucial by supervivors as in Solvency II (European Insurance and Occupational Pensions Authority [EIOPA] 2009) as well as in International Financial Report Standard no. 17 (IFRS 2017), hereafter IFRS 17. Risk margin assessment is largely debated by both practitioners and academic actuaries because there is not a single definition or a unique model for the related estimation; for an extended debate see Risk Margin Working Group of the International Actuarial Association (2009). Risk adjustment is the compensation that the entity requires for bearing uncertainty about the amount and timing of cash flows that arise from nonfinancial risk and, for example, IFRS 17 does not specify the technique to determine the risk adjustment, but it requires a disclosure of confidence level required irrespective of the technique used to measure it. Furthermore, in Risk Margins Taskforce (1998), several methods have been set forth for risk margin estimation involving the simple analysis of the underlying risks or a wider range of information and techniques such as stochastic modeling. The assessment of the risk margin could be evaluated by means of a specific risk measure applied to a loss (or profit) random variable (r.v.). The first use of

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a risk measure in actuarial science was the development of premium calculation principles—as mentioned in the classic actuarial literature (Goovaerts, De Vylder, and Haezendonck 1984; Hardy 2006)—and were essentially based on variability measures of the loss.

With reference to non-life insurance, also called general insurance or property and casualty insurance, the process of establishing premiums (or rates) is also known as a ratemaking process. Ratemaking is a typical actuarial problem and its goal is essentially to ensure that premiums can cover all expected costs. It is performed on a collective level (e.g., line of business, product) as well as at an individual or segment level (e.g., insured, risk class). Ratemaking of non-life individual insurance risk is frequently performed by means of multiple regression models that allow the investigation of several rating factors (see Werner and Modlin 2016).

In this article we introduce a premium principle based on a quantile risk measure, in order to investigate the risk margin in the ratemaking process at individual level. We start by considering a traditional two-part (also known as two-stage) model (Duan et al. 1983) to decompose the cost of a claim between the event of the claim and its amount if the claim occurs. Then we define the risk margin considering a quantile of the claim's severity, for each individual risk class. To this aim we compare two multiple regression models: generalized linear models (GLMs) and quantile regression (QR). By means of multiple regression models, it is possible to calculate a risk margin via quantile for each individual premium, taking into account those risk factors that best characterize the risk of each policyholder at the chosen probability level.

GLMs were introduced by Nelder and McCullagh (1989) and are considered the widely recognized industry standard method for pricing many lines of business in many countries (see Werner and Modlin 2016). It is worth noting that in order to analyze the variability of the loss distribution, GLMs permit defining the conditional distribution considering that the parameters are always well defined.

So, our first contribution is to assess the risk margin for each risk profile using a quantile approach via a two-stage gamma GLM. The gamma distribution, among other distributions belonging to the exponential family, is a natural fit as well as the most widely used in practice for modeling severity in a GLM (see Goldburd, Khare, and Tevet 2016). Its characteristics as right-skewed, with a sharp peak and a long tail to the right, make it suitable to model claim severity because these features are usually exhibited by empirical claims' distribution. We demonstrate analytically that, under this assumption, the premium model we proposed implies a risk margin proportional to the expected value identical for each risk class and is only dependent on the selected probability level.

As an alternative, considering the well-known drawbacks of GLMs in assessing higher distribution moments and the high sensitivity to the outlier, we propose a ratemaking approach based on a two-stage QR (Biancalana 2017; Heras, Moreno, and Vilar-Zanón 2018). QR was introduced by Koenker and Bassett in 1978 and it is capable of providing a more robust statistical analysis than ordinary least squares (OLS) estimation, when the least squares model assumptions are rejected. Moreover, it is a distribution-free method because it does not need any assumptions about a parametric distribution family or use any properties of it, it is less sensitive to data outliers, and extreme data values do not influence the regression results as significantly as in GLMs.

QR is widely applied in several fields ranging from finance to survival analysis but has rarely been applied in actuarial science.

The use of QR for ratemaking has been discussed in the article by Kudryavtsev (2009), in the working paper by Wolny, Ornat-Acedaska, and Trzpiot (2012), in the Ph.D thesis by Biancalana (2017), and in the paper by Heras, Moreno, and Vilar-Zanón (2018). The last two works propose a two-stage QR model based on a logistic for the frequency and a QR for severity for the individual premium calculation and can be considered a refinement of the model proposed by Kudryavtsev (2009) where a single probability of having claims for the different policies has been assumed. The approach suggested in Heras, Moreno, and Vilar-Zanón is based on the definition of a single level of probability to estimate the aggregate claim amount for each risk class. However, this approach is arguable because it involves some practical issues when applied to large portfolios (e.g., motor third party liability, MTPL); indeed, if it is applied in portfolios with a large number of risk classes, it requires the calibration of a number of QRs equal to the risk classes considered that can exceed thousands of units in some real data-set.

Hence, in order to overcome these drawbacks, our second contribution is to propose an alternative approach based on a two-stage QR, as introduced in Biancalana (2017), where the risk margin is calibrated on the claim's severity for each risk class in the portfolio: in this way, we prevent the need to calibrate different QRs for each risk class and we avoid the subsequent over parameterization.

Finally, our method permits estimating premium rates considering a unique probability at the individual level for the quantile estimate. The probability level is defined such that the equilibrium at the aggregate level between the total premium for all policies written is sufficient to cover the total expected losses and expenses and to provide for the targeted profit or to respect solvency constraints.

The rest of the article is organized as follows. Section 2 introduces the premium calculation principle and the decomposition of the loss into a two-part model consisting of frequency (number) and severity (amount) components. Section 3 illustrates the quantile pricing model we suggest for the assessment of risk margin and the solution provided in both two-stage gamma GLM and QR cases. Section 4 presents the case study concerning the risk margin assessment using a real dataset and illustrates the results. Finally, Section 5 discusses the main findings and presents conclusions.

2. ACTUARIAL PREMIUM PRINCIPLES AND THE TWO-PART MODEL FOR NON-LIFE INSURANCE

One of the main tasks of a pricing actuary is to determine rates that will produce a premium in balance with the expected costs (i.e., losses and expenses) over time and include a profit margin at both individual or aggregate levels. The calculation methods adopted to define premium rates in life and non-life insurance sectors are usually referred to as premium calculation principles. Premium principles are included in the broadest class of risk measures and can be interpreted as both premiums or capital requirements for insurance risks. We define a set of random variable Υ in the probability space $[\Omega, \Im, \mathbf{P}]$. The elements $Y \in \Upsilon$ are nonnegative random variables, representing the risks the insurance company is exposed to at a fixed time horizon T. Let F_Y be the cumulative density function (cdf) of Y. If we consider a non-life insurance undertaking, then generally $T = 1$. If for $\omega \in \Omega, Y(\omega) > 0$ then we will consider this to be a loss. A risk measure is a functional π that assigns a non negative real number to random variables Y representing uncertain pay offs, in our context the insurance losses.

$$\pi(Y) : \Upsilon \rightarrow \mathbb{R}. \tag{2.1}$$

Moreover, we assume that the expected value $E(Y)$, the variance $Var(Y)$, and the moment-generating function $M_Y(z) = E(e^{zY})$ exist.

The simplest calculation principle is based on the *equivalence* premium principle. The net premium is equal to the expectation of loss $E(Y)$. As known from ruin theory, it is simple to show that if an insurer adopts the equivalence premium principle, this will lead to his or her ruin with certainty. So, to overlap this issue, it is necessary to include a positive "safety loading." Traditional "loaded" net premium principles correct the equivalence principle in order to generate a premium that is grater than the expected loss. The difference between the premium and the expected loss is called the "premium loading." A survey of premium calculation principles may be found in Bowers et al. (1989), Bühlmann (1970), Gerber (1979), and Laeven and Goovaerts (2008).

For practical purposes, it is common to express the premium loading as a percentage of the expected value; that is,

$$\rho(Y) = \frac{\pi(Y)}{E(Y)}. \tag{2.2}$$

Hereafter, we call $\rho(Y)$ the "risk margin function."

It is worth noting that, regardless of the premium principle adopted, the assignment of a real number through the mapping of $\pi(Y)$ can be expressed using the inverse of the cdf of Y —that is, the quantile function $Q_\theta(Y)$, if it exists, as follows:

$$\pi(Y) = F_Y^{-1}(\theta) = Q_\theta(Y) \quad , \theta \in [0, 1]. \tag{2.3}$$

For continuous distributions, the quantile function Q returns the value $y = Q_\theta$ such that

$$F_Y(Q_\theta) := Prob(Y \leq Q_\theta) = \theta. \tag{2.4}$$

In other words, it is always possible to define the premium as the θ-quantile of the loss distribution; this approach is also known as the *quantile* premium principle.

The basic *quantile* premium principle is based on a *Value at Risk* (VaR) measure widely applied in finance and crucial in most VaR-based risk management systems in both financial and insurance sectors as in Basel II and Solvency 2. For example, in the insurance sector, Solvency 2 establishes that premium risk will be evaluated in order to consider for each segment (e.g., MTPL, General Third Party Liability (GTPL), fire, etc.) a solvency capital requirement corresponding to the VaR subject to a confidence level of 99.5% over a one-year period; so, in the intention of the European supervisor, it seems admissible to write $\pi(Y) = Q_{99.5}(Y)$.

The problem of spreading the overall premium over the individual risks (i.e., policies) is addressed in different ways and involves the concept of risk segmentation via rating factors. This last process in non-life insurance is known as classification

ratemaking (see Werner and Modlin 2016). In the following, we shall consider a non-life insurance undertaking whose portfolio r of insurance contracts is composed by i homogeneous individual risks. In an individual risk model the loss of the i th individual risk is modeled by means of the generic random variable $Y^{(i)} \in \Upsilon, i = 1, 2, ..., r$ representing the claim amount per the i th insured. The random variables $Y^{(i)}$ are assumed to be independent and their sum is the portfolio's total loss Y.

$$Y = \sum_{i=1}^{r} Y^{(i)} = \sum_{i=1}^{r} \mathbb{I}_{N^{(i)}} \ \tilde{Y}^{(i)}, \tag{2.5}$$

with $\mathbb{I}_{N^{(i)}}$ be an indicator random variable indicating whether the i th insured has had at least one claim; $\tilde{Y}^{(i)} = (Y^{(i)}|N^{(i)} > 0)$ is a strictly positive random variable representing the total loss per claimant, with $N^{(i)}$ the random number of claims of the i th insured.

The indicator random variable $\mathbb{I}_{N^{(i)}}$ is generally Bernoulli (p_i) distributed and independent of $\tilde{Y}^{(i)}$.

Considering that $Prob[Y^{(i)} \leq 0|N^{(i)} = 0] = 1$, the cdf $F_{Y^{(i)}}(y)$ can be written as

$$\begin{aligned} &Prob\left[Y^{(i)} \leq y\right] = Prob\left[Y^{(i)} \leq y, \mathbb{I}_{N^{(i)}} = 0\right] + Prob\left[Y^{(i)} \leq y, \mathbb{I}_{N^{(i)}} = 1\right] = \\ &p_i + (1-p_i) Prob\left[\tilde{Y}^{(i)} \leq y\right]. \end{aligned} \tag{2.6}$$

In actuarial literature, this way of decomposing the total claim amounts is traditionally referred to as a frequency–severity model. To predict the cost of a claim, it is a usual approach to decompose two-part data: one part for the frequency, indicating whether a claim has occurred or, more generally, the number of claims, and one part for the severity, indicating the amount of a claim. In statistics, such models are generally called two-part models (Duan et al. 1983).

Beyond theoretical considerations, even from a practical point of view, many insurers keep separate data files for the policyholder characteristics (e.g., gender, age, car type, etc.) and insurance claim events (e.g., accident date, severity, number of bodily injuries, etc.). This recording process makes it natural for insurers to model the frequency and severity as separate processes.

Moreover, depending on the risk being analyzed, insured datasets may contain a large proportion of zeros–as in MTPL insurance–that must be accounted for in the modeling. A two-part model is based on observing a dependent variable, such as a claim, where it is known that it cannot be less than zero. For a discussion on the advantages of a frequency–severity approach for insurance rating, see Goldburd, Khare, and Tevet (2016).

3. A QUANTILE PREMIUM PRINCIPLE BASED ON A TWO-PART MODEL

In order to assess the pure premium with the risk model described in Section 2, it is common to develop a separate analysis of the factors or explanatory variables that influence frequency and severity, due to their independence. Then, the individual models are combined to form a pure premium model.

In actuarial practice, this analysis is implemented through multiple regression techniques such as GLMs assuming a link for the response variables, generally a log or logit function, in order to obtain a multiplicative tariff. As stated before, the combination of the two parts is achieved by simply multiplying their corresponding relativity factors together. The advantage of GLM is in the ability to estimate the conditional expected value of the response variable. Consequently, the estimate of the equivalence premium $E(Y^{(i)})$ is

$$E\left(Y^{(i)}\right) = E\left(\mathbb{I}_{N^{(i)}}\right) \cdot E\left(\tilde{Y}^{(i)}\right). \tag{3.1}$$

Moreover, to assess the risk margin, we need to estimate moments of $Y^{(i)}$ other than the mean. The solution we propose is to adopt a count regression model for the estimation of the conditional expectation of the frequency component and a fat-tailed regression model in order to assess the conditional quantile of the severity component as proposed in Biancalana (2017) and Heras, Moreno, and Vilar-Zanón (2018).

Therefore, we shall focus on the assessment of the quantile of $\tilde{Y}^{(i)}$, and the solution we suggest is to adopt a pure premium defined as follows:

$$P_Q = E\left(\mathbb{I}_{N^{(i)}}\right) \cdot Q_{\tilde{\theta}_i}\left(\tilde{Y}^{(i)}\right) \tag{3.2}$$

such that the risk margin functions is

$$\rho\left(Y^{(i)}\right) = \frac{P_Q}{E\left(Y^{(i)}\right)} = \frac{Q_{\tilde{\theta}_i}\left(\tilde{Y}^{(i)}\right)}{E\left(\tilde{Y}^{(i)}\right)} = \rho\left(\tilde{Y}^{(i)}\right). \tag{3.3}$$

It is worth noting that the use of a different functional for frequency and severity, the expectation and the quantile respectively, could be arguable. In fact, a different solution such as a quantile–quantile approach may be addressed. However, this choice is equally arguable because it presents the following drawbacks: in (3.2) the application of a quantile function to the indicator random variable $\mathbb{I}_{N^{(i)}}$ is useless because quantiles assume only a value of 0 or 1.

Let $Q_{\tilde{\theta}_i}$ be the $\tilde{\theta}_i$th quantile of the truncated or conditional random variable loss incurred for the i th claimant, $\tilde{Y}^{(i)}$, where it is assumed that the number of claims is strictly positive. In order to define the distribution of loss incurred for the i th insured, $Y^{(i)}$, the event "no claims" should be considered. Hence, let $p_i = Prob(N^{(i)} = 0)$ be the probability of no claims. By Equation (2.6) it holds that

$$F_{Y^{(i)}}(Q_{\tilde{\theta}_i}) = p_i + [1 - p_i] \cdot F_{\tilde{Y}^{(i)}}(Q_{\tilde{\theta}_i}) \tag{3.4}$$

$$F_{Y^{(i)}}(Q_{\tilde{\theta}_i}) = p_i + [1 - p_i] \cdot \tilde{\theta}_i. \tag{3.5}$$

Let $F_{Y^{(i)}}(Q_{\tilde{\theta}_i}) = \theta_i$, then

$$\theta_i = p_i + [1 - p_i] \cdot \tilde{\theta}_i \quad \Longleftrightarrow \quad \tilde{\theta}_i = \frac{\theta_i - p_i}{1 - p_i}. \tag{3.6}$$

Equation (3.6) shows a relation between $\tilde{\theta}_i$ and θ_i. Moreover, by (2.6) and (3.6) we have that

$$Q_{\theta_i}\left(Y^{(i)}\right) = Q_{\tilde{\theta}_i}\left(\tilde{Y}^{(i)}\right). \tag{3.7}$$

In the actuarial literature, Kudryavtsev (2009) was the first to use QR for premium calculation, but he considered a constant probability of claim $p_i \equiv p, i = 1, ..., r$ for each insured. An improvement of this approach was proposed in Biancalana (2017) and Heras, Moreno, and Vilar-Zanón (2018) where a different claims probability for each risk class was considered and modeled using a GLM.

Moreover, in order to assess the premium rates, Heras, Moreno, and Vilar-Zanón (2018) suggested fixing a unique probability level θ (in their example equal to 99.5%) for the individual aggregate amount $Y^{(i)}$; then, by means of (3.6) it is possible to define the corresponding set of different $\tilde{\theta}_i, i = 1, ..., r$ for the severity amount $\tilde{Y}^{(i)}$. However, for calculation of premium rates it is necessary to run a set r of QR models, with each one calibrated at a different probability level $\tilde{\theta}_i$ associated with the relative i th risk class. Although the approach is robust from a theoretical point of view, it has practical issues when the number of risk classes is high as in a real dataset. Indeed, when performed on large portfolios, it implies an excessive cost in terms of model validation as well as an overparameterization.

Thus, in order to overcome these drawbacks, we suggest the following approach. We fix a unique quantile for the severity $\tilde{Y}^{(i)}$—that is, $\tilde{\theta}_i \equiv \tilde{\theta}$—and then the corresponding θ_i probability level of $Y^{(i)}$ obtained via Equation (3.6) will depend for each insured only on p_i and $\tilde{\theta}$.

Hence, Equation (3.2) is simply obtained as

$$P_Q^{(i)} = (1 - p_i) \cdot Q_{\tilde{\theta}}\left(\tilde{Y}^{(i)}\right). \tag{3.8}$$

According to (3.8), our next step consists of the estimate of the two components of the net premium, the probability of the event of the claim $(1-p_i)$ and the $\tilde{\theta}$-quantile of the severity $Q_{\tilde{\theta}}(\tilde{Y}^{(i)})$. The former can be addressed by a traditional logit or probit GLM. The latter requires a slightly more complex approach that we will examine in the following subsections. In particular, we shall investigate a gamma GLM with unitary prior weights (i.e., $\omega_i = 1, i = 1, 2, ..., r$) and a QR approach respectively (see Appendixes A and B for further details).

TABLE 1
Characteristics of Some Common Univariate Distributions in the Exponential Family

Density function	Dispersion parameter	Canonical parameter	Mean value	Variance function
$f(y,\kappa,\varphi)$	φ	κ	$\mu(\kappa)$	$Var(\mu)$
$\frac{1}{\Gamma(\nu)y}(\frac{y}{\eta})^{\nu}e^{-\frac{y}{\eta}}$	$\frac{1}{\nu}$	$-\frac{1}{\eta\nu}$	$-\frac{1}{\kappa}$	μ^2

3.1. Risk Margin Estimation via Gamma GLM: The Case with Unitary Prior Weights

Let us consider estimating the conditional quantile of the severity of each claimant $\tilde{Y}^{(i)}, i = 1, 2, ..., r$ by means of a gamma GLM. The underlying assumption of the gamma GLM is that the response variable is gamma distributed, that is, $\tilde{Y}^{(i)} \sim G(\nu_i, \eta_i)$ with the following characteristics[1] (see Table 1):

Nevertheless, if we consider that each claimant has the same information weight, this implies considering prior weight ω_i equal to 1 for each $i = 1, 2, ..., r$; thus, we can state that the use of a gamma GLM leads to the following conditions:

Proposition 1: The estimated coefficient of variation (CV) of each individual severity per claimant is constant and equal to the square root of the dispersion parameter φ. Then, the estimated distribution of $\tilde{Y}^{(i)}$ is characterized by a unique shape parameter $\nu_i \equiv \nu$ for each $i = 1, ..., r$.

Proposition 2: The $\tilde{\theta}$ th quantile $Q_{\tilde{\theta}}(\tilde{Y}^{(i)})$ is obtained as a product of the expected value $E(\tilde{Y}^{(i)})$ and a constant depending exclusively on level $\tilde{\theta}$, say $\rho_{\tilde{\theta}}$. As a consequence, the risk margin function in Equation (2.2) is the same for each insured; that is, $\rho_{\tilde{\theta}}(\tilde{Y}^{(i)}) = \rho_{\tilde{\theta}}$. Thus,

$$Q_{\tilde{\theta}}\left(\tilde{Y}^{(i)}\right) = \rho_{\tilde{\theta}} \cdot E\left(\tilde{Y}^{(i)}\right) = \rho_{\tilde{\theta}} \cdot \eta_i \cdot \nu. \tag{3.9}$$

Proof. Proposition 1

As $\omega_i \equiv 1$ for each $i = 1, 2, ..., r$, the function $a(\varphi)$ assumes a value equal to φ.

Therefore, by Equation (A.2) the variance is expressed as follows:

$$Var(\tilde{Y}_i) = \varphi \cdot \mu_i^2 = \varphi \cdot (\eta_i \cdot \nu_i)^2 . \tag{3.10}$$

Then, the CV is the same for each $\tilde{Y}^{(i)}$ as

$$CV\left(\tilde{Y}^{(i)}\right) = \sqrt{\frac{1}{\nu_i}} = \frac{\sigma\left(\tilde{Y}^{(i)}\right)}{E\left(\tilde{Y}^{(i)}\right)} = \frac{\sqrt{\varphi \cdot \mu_i^2}}{\mu_i} = \sqrt{\varphi}, i = 1, ..., r. \tag{3.11}$$

Therefore, it implies that $\nu_i \equiv \nu \equiv 1/\varphi$. □

Proof. Proposition 2

As proved in Proposition 1, the r.v. $\tilde{Y}^{(i)} \sim G(\nu, \eta_i)$ is gamma distributed with a unique shape parameter ν. From Equation (2.4):

$$\tilde{\theta} = Prob\left\{\tilde{Y}^{(i)} < Q_{\tilde{\theta}}\left(\tilde{Y}^{(i)}\right)\right\} = \int_0^{\rho_{\tilde{\theta}} \cdot \nu \cdot \eta_i} \frac{1}{\Gamma(\nu)\eta_i^{\nu}} x^{\nu-1} e^{-\frac{x}{\eta_i}} dx. \tag{3.12}$$

[1]The parameterization of the gamma distribution is such that its variance is $\nu_i \cdot \eta_i^2$.

Let $r = 2$; if ***Proposition 2*** holds true, it implies by Equation (3.9) that

$$\tilde{\theta} = Prob\{\tilde{Y}^{(1)} < \rho_{\tilde{\theta}} \cdot \nu \cdot \eta_1\} = Prob\{\tilde{Y}^{(2)} < \rho_{\tilde{\theta}} \cdot \nu \cdot \eta_2\}.$$

Then

$$\int_0^{\rho_{\tilde{\theta}} \cdot \nu \cdot \eta_1} \frac{1}{\Gamma(\nu)\eta_1^{\nu}} x^{\nu-1} e^{-x/\eta_1} dx = \int_0^{\rho_{\tilde{\theta}} \cdot \nu \cdot \eta_2} \frac{1}{\Gamma(\nu)\eta_2^{\nu}} x^{\nu-1} e^{-x/\eta_2} dx \Longleftrightarrow$$

$$\eta_2^{\nu} \int_0^{\rho_{\tilde{\theta}} \cdot \nu \cdot \eta_1} x^{\nu-1} e^{-x/\eta_1} dx \quad = \eta_1^{\nu} \int_0^{\rho_{\tilde{\theta}} \cdot \nu \cdot \eta_2} x^{\nu-1} e^{-x/\eta_2} dx. \quad .$$

By deriving the previous expression with respect to $\rho_{\tilde{\theta}}$, the equality holds true:

$$\frac{d}{d\rho_{\tilde{\theta}}} \left[\eta_2^{\nu} \int_0^{\rho_{\tilde{\theta}} \cdot \nu \cdot \eta_1} x^{\nu-1} e^{-x/\eta_1} dx \right] = \frac{d}{d\rho_{\tilde{\theta}}} \left[\eta_1^{\nu} \int_0^{\rho_{\tilde{\theta}} \cdot \nu \cdot \eta_2} x^{\nu-1} e^{-x/\eta_2} dx \right] \Longleftrightarrow$$

$$\nu \cdot \eta_1 \cdot \eta_2^{\nu} (\rho_{\tilde{\theta}} \cdot \nu \cdot \eta_1)^{(\nu-1)} e^{-\rho_{\tilde{\theta}} \cdot \nu} = \nu \cdot \eta_1^{\nu} \eta_2 (\rho_{\tilde{\theta}} \cdot \nu \cdot \eta_2)^{(\nu-1)} e^{-\rho_{\tilde{\theta}} \cdot \nu}.$$

By Equation (3.8), if we adopt a gamma GLM for $\tilde{Y}^{(i)}$, we have

$$P_{Q_{GLM}}^{(i)} = [1 - p_i] \cdot \rho_{\tilde{\theta}} \cdot E\left(\tilde{Y}^{(i)}\right) = \rho_{\tilde{\theta}} \cdot E\left(Y^{(i)}\right) \tag{3.13}$$

and the risk margin function is

$$\rho_{GLM}\left(Y^{(i)}\right) = \frac{P_{Q_{GLM}}^{(i)}}{E\left(Y^{(i)}\right)} = \rho_{\tilde{\theta}}. \tag{3.14}$$

In a nutshell, the risk margin estimation via a two-part model based on a gamma GLM for the severity per claimant with prior weights equal to one, leads to the following drawbacks:

- The assumption of an independent gamma distribution for each insured i with the same CV is arguable and, if not confirmed by data, leads to inconsistent quantile estimation.
- If we adopt a quantile premium calculation principle, the risk margin function is the same (3.14) for each risk class i and the premium loading depends exclusively on the expected value of $Y^{(i)}$.
- Then the solution found is equivalent to the application of the *expected value* premium principle because it simply implies that $P_Q^{(i)} = (1 + \alpha) \cdot E(Y^{(i)}) = \rho_{\tilde{\theta}} \cdot E(Y^{(i)})$ with a fixed loading factor equal for each risk class.

3.2. Risk Margin Estimation with QR

In order to overcome the previously stated drawback of gamma GLMs', we shall suggest, as an alternative, using QR for the estimation of $Q(Y^{(i)})$, which enables a better fitting of the conditional quantile than GLMs. The application of QR as suggested in Equation (3.8) requires the combination of the two-part model by multiplying the estimates obtained with the logit or probit model for the frequency and those obtained with QR for the severity. Therefore, to be compliant with a multiplicative tariff, the log transformation of the response variables adopted for GLMs (see Equation [A.4]) have been adopted for QR through the "equivariance to monotone transformation." Indeed, if the response variable is transformed by logarithm, then by Equarion (B.4) we have

$$Q_{\tilde{\theta}}\left(log\left(\tilde{Y}^{(i)}\right)\right) = log\left(Q_{\tilde{\theta}}\left(\tilde{Y}^{(i)}\right)\right) = \mathbf{x}_i \cdot \beta_{\tilde{\theta}} \rightarrow Q_{\tilde{\theta}}\left(\tilde{Y}^{(i)}\right) = e^{\mathbf{x}_i \cdot \beta_{\tilde{\theta}}}. \tag{3.15}$$

As in the GLM case, the QR leads to a different quantile level of $Y^{(i)}$ for each risk class and consequently to different premium loadings as expressed in Equation (3.8). However, the risk margin function (2.2) in the QR case is not equal. Indeed, by Equations (A.4), (3.8), and (3.15), it holds that

$$\begin{aligned}\rho_{QR}\left(Y^{(i)}\right) &= \frac{P_{QR}^{(i)}}{E\left(Y^{(i)}\right)} = \frac{(1-p_i)\cdot Q_{\theta_i}\left(Y^{(i)}\right)}{E\left(Y^{(i)}\right)} = \\ &= \frac{(1-p_i)\cdot Q_{\tilde{\theta}}\left(\tilde{Y}^{(i)}\right)}{(1-p_i)\cdot E\left(\tilde{Y}^{(i)}\right)} = \frac{Q_{\tilde{\theta}}\left(\tilde{Y}^{(i)}\right)}{E\left(\tilde{Y}^{(i)}\right)} = \frac{e^{\mathbf{x}_i\cdot\beta_{\tilde{\theta}}}}{e^{\mathbf{x}_i\cdot\beta}} = e^{\mathbf{x}_i\cdot(\beta_{\tilde{\theta}}-\beta)\cdot}\end{aligned} \tag{3.16}$$

The above suggested solution put in evidence the pure dependency of the risk function on the risk class characteristics included in the row vector of independent covariates, $\mathbf{x}_i = (x_{i1}, ..., x_{im})$. So, compared to the solution provided with a gamma GLM (see Equation [3.14]), the risk function QR allows for a better description of the probability distribution of each risk class considered.

3.3. Calibration of the Individual Premium Security Level

Once the model for the individual risk premium has been defined, it is necessary to determine the probability level to be applied for pure premium calculation. The basic idea is that the sum of all premiums covers the expected aggregate amount of loss plus a risk margin value. The overall aggregate amount is generally selected by insurance managers by looking at pricing or solvency conditions, so one possible choice is to calibrate the individual probability level $\tilde{\theta}$ such that the balance equation is satisfied:

$$\pi(Y) = \sum_{i=1}^{r} P_Q^{(i)} = \sum_{i=1}^{r}[1-p_i]\cdot Q_{\tilde{\theta}}\left(\tilde{Y}^{(i)}\right). \tag{3.17}$$

The calibration procedure consists of finding the security level $\tilde{\theta}^*$ such that the overall risk premium is spread over each individual risk; it is worth noting that, the solution proposed permits allocating the overall risk margin over the policyholders considering their specific risk preference.

As an example, if we adopt the approach suggested by EIOPA (2009) for risk premium calculation, we have $\pi(Y) = Q_{99.5}(Y)$; hence, the overall risk margin function is $\rho(Y) = \frac{Q_{99.5}(Y)}{E(Y)} = 1+\alpha$.

It is easy to demonstrate that if we adopt the two-stage gamma GLM by Equation. 3.14 we have

$$\rho(Y) = \frac{Q_{99.5}(Y)}{E(Y)} = \frac{\sum_{i=1}^{r} P_{Q_{GLM}}^{(i)}}{\sum_{i=1}^{r} E(Y^{(i)})} = \frac{\sum_{i=1}^{r}[1-p_i]\cdot\rho_{\tilde{\theta}}\cdot E\left(\tilde{Y}^{(i)}\right)}{\sum_{i=1}^{r}[1-p_i]\cdot E\left(\tilde{Y}^{(i)}\right)} = \rho_{\tilde{\theta}} = (1+\alpha). \tag{3.18}$$

Because we calculate quantiles via GLM and QR, before the pure premium estimation we have to solve (3.17) for the two regression models. For the gamma GLM, given that $\rho(Y) = \rho_{\tilde{\theta}}$, one should solve iteratively the following equation for $\tilde{\theta}^*_{GLM}$, where ν, η_i are the parameters of the gamma distribution of the generic i th risk class:

$$\tilde{\theta}^*_{GLM} = \int_0^{(1+\alpha)\cdot\nu\cdot\eta_i} \frac{1}{\Gamma(\nu)\eta_i^{\nu}} x^{\nu-1} e^{-\frac{x}{\eta_i}} dx. \tag{3.19}$$

For the QR model, it is necessary to improve an iterative method at a different probability level $\tilde{\theta}$ in order to find the $\tilde{\theta}^*_{QR}$ that better approximates Equation (3.17).

It is worth noting that the solutions $\tilde{\theta}^*_{QR}$ and $\tilde{\theta}^*_{GLM}$ could be different, as we demonstrate empirically in the next section.

TABLE 2
Dataset Rating Factors

Rating factors	Levels
Age band of policy holder	1 (youngest), 2, 3, 4, 5, 6
Gender	M and F
Area of residence	A, B, C, D, E, F
Vehicle age	1 (new), 2, 3, 4
Vehicle body type	Bus, convertible, coupe, hatchback, hardtop, motorized caravan combi, minibus, panel van, roadster, sedan, station wagon, truck, utility vehicle

TABLE 3
Risk Classes—Descriptive Statistics

h	Risk class	No. of policies	Exposure	No. of claimed policies	Average claim severity	Coefficient of variation
1	1-F	7,145	3,084.20	482	1,677.49	2.22
2	2-F	9,935	4,740.18	768	1,790.14	1.63
3	3-F	11,707	5,504.78	783	1,891.06	1.46
4	4-F	9,816	4,625.44	615	2,023.85	1.40
5	1-M	5,112	2,254.75	343	2,176.99	2.09
6	2-M	6,652	3,183.50	491	2,263.52	2.14
7	3-M	8,357	4,037.33	579	2,137.36	1.65
8	4-M	9,132	4,370.64	563	2,327.23	1.64
Total		67,856	31,800.82	4,624	2,104.40	1.71

4. NUMERICAL APPLICATION

In order to conduct a comparison between the previously described conditional quantile estimation approaches, we present an application to a real insurer database. Our aim is to analyze the ability of the proposed two-stage models to predict the risk class probability distribution, specifically focusing on the risk margin value measured by a selected quantile.

4.1. Dataset

We demonstrate our model using database from an Australian automobile insurance company (De Jong and Heller 2008) between 2004 and 2005, which is available in the R package "Insurancedata." The portfolio has 67,856 policies, with 4624 policies with at least one claim. The rating factors in this dataset are listed in Table 2.

Having the goal of a full ratemaking process, it is worth noting that it would be necessary to compute 3744 premium rates, one for each risk class obtained as the combination of the levels assumed by the five rating factors.

To simplify our representation, in the following we shall consider only two rating factors: vehicle age (hereafter Veh_{Age}) and *Gender*; in this way, we limit our analysis to eight rating classes obtained by the combination of four levels $(1, 2, 3, 4)$ and two levels (M, F) for Veh_{Age} and *Gender,* respectively.

Some synthetic descriptive statistics of risk classes considered are provided in Table 3.

Let:

- $r = 67,856$ elements set of policies;
- $i \in \mathbb{G}$ index r elements set;
- $h = 1, \ldots, 8$ index the number of risk classes;
- $\mathbb{D}_h \subseteq \mathbb{G}$ is the g_h-elements set of exposure belonging to the h th risk class, where $\sum_{h=1}^{8} g_h = 67,856$;

- $\mathbb{C}_h \subseteq \mathbb{D}_h \subseteq \mathbb{G}$ is the c_h-elements set of claimants belonging to the h th risk class, where $\sum_{h=1}^{8} c_h = 4624$;
- $y^{(i\in\mathbb{D}_h)}$ is the loss generated for the i th exposure belonging to the h th risk class,

$$y^{(i\in\mathbb{D}_h)} = \begin{cases} 0 & \text{if } i \in \bar{\mathbb{C}}_h \\ \tilde{y}^{(i\in\mathbb{C}_h)} & \text{if } i \in \mathbb{C}_h \end{cases},$$

where $\tilde{y}^{(i\in\mathbb{C}_h)}$ represents the positive loss associated with the i th claimant belonging to $\mathbb{C}_h$.

- $\mathbf{x}_{i\in\mathbb{D}_h}$ is the row vector of the design matrix, providing information about exposure rating factors. Thus, we can state $\mathbf{x}_{i\in\mathbb{D}_h} = \mathbf{x}_h$ for all $i \in D_h$.

Therefore, the aim is the estimate of

$$P_Q\left(Y^{(i\in\mathbb{D}_h)}\right) = P_Q(Y|x_{i\in\mathbb{D}_h}) = P_Q(Y|x_h) = P_Q\left(Y^{(h)}\right). \tag{4.1}$$

4.2. GLM Models for Conditional Mean Estimations

The first step concerns the estimate of $Prob[N^{(h)} > 0] = (1-p_h)$. The latter is performed by a GLM binomial logit function. The model leads to the following estimate:

$$E[I_{(i\in C_h)}] = (1 - p_h) = g^{-1}(x_h \cdot \gamma), \tag{4.2}$$

where $g(.)$ is the logit link function and

$$I_{(i\in C_h)} \begin{cases} 1 & \text{if } i \in C_h \\ 0 & \text{if } i \in \bar{C}_h \end{cases}.$$

The parameter estimates for the regression models used herein are obtained by using maximum likelihood estimation and are reported in Table 4.

As one can observe, regression parameters $Veh_{Age} = 2$ and $Gender = M$ are of low significance and levels could be aggregated to the base lever. However, for our analysis we leave the levels separated.

Then, the second step concerns the estimate of $E(\tilde{Y}^{(i)})$ through a gamma GLM with log link function. The model leads to the parameter estimates reported in Table 5:

Regression parameters $Veh_{Age} =$ 2,3, and 4 are of low significance and levels should be aggregated to the base level. However, as previously stated, we leave the levels separated.

Finally, in Table 6 we show the estimates of the conditional mean of the probability of no claim p_h and claim severity per claimant $E(\tilde{Y}^{(h)})$. Moreover, we report the equivalence premium for each risk class obtained by means of (3.1).

It is worth noting that the application of the equivalence premium permits a ranking in terms of expected values showing that, based on this principle, the "riskiest" risk class is 2-M and the least risky is 4-F. It is obvious that the equivalence premium is not a risk-based approach because no risk measure has been considered. In the following we will consider a quantile risk measure that could subvert the ranking obtained with the principle discussed here.

TABLE 4
Estimates of Parameters for Frequency Model

Parameter	Value	SE	p Value
$\gamma_0, Intercept$	–1.689	0.040	
$\gamma_1, Veh_{Age} = 2$	0.026	0.049	.590
$\gamma_2, Veh_{Age} = 3$	–0.099	0.048	.039
$\gamma_3, Veh_{Age} = 4$	–0.200	0.049	$<10^{-3}$
$\gamma_4, Gender = M$	–0.027	0.032	.590

TABLE 5
Estimates of Parameter for Claim Severity Conditional Mean

Parameter	Value	SE	p Value
β_0, *Intercept*	7.459	0.065	
β_1, Veh_{Age}=2	0.052	0.078	.504
β_2, Veh_{Age}=3	0.061	0.077	.425
β_3, Veh_{Age}=4	0.134	0.079	.091
β_4, *Gender* Male	0.182	0.052	.000

TABLE 6
Conditional Mean Estimates and Equivalence Premium

h	Risk class	Exposure	p_h	$E(\tilde{Y}^{(h)})$	$E(Y^{(h)})$
1	1-F	3084.20	0.844	1734.85	270.45
2	2-F	4740.18	0.841	1828.16	291.38
3	3-F	5504.78	0.857	1844.88	264.42
4	4-F	4625.44	0.869	1984.13	260.67
5	1-M	2254.75	0.848	2080.32	316.95
6	2-M	3183.50	0.844	2192.21	341.51
7	3-M	4037.33	0.860	2212.26	309.78
8	4-M	4370.64	0.872	2379.24	305.28

4.3. QR Models for Conditional Quantile Estimations

The third step consists of the estimation of $Q_{\tilde{\theta}}(\tilde{Y}^{(h)})$.

The parameter estimates for the QR model are obtained by using least absolute deviations (B.2) because QR is a distribution-free approach. The following figures show the estimates of the parameters. The base class or intercept is represented by the 1-F risk class. It is worth noting that for QR we have performed a set of QR models for each element of the vector QR with $\tilde{\theta}=(0.01, 0.02, ..., 0.99)$, in order to discuss the behavior of the regression parameters with respect to the probability level. It is worth noting that this analysis is preparatory for the next steps; though it is not strictly necessary, it is helpful.

Figure 1 shows a basic representation of the estimated parameters with the QR against gamma GLM and allows a comparison over the entire range of probability levels. In each graph, the level of probability $\tilde{\theta}$ is shown on the x-axis, and the y-axis shows that parameter estimates. The black points represent QR coefficients estimated according to different probability levels $\tilde{\theta}$. The gray band represents confidence intervals at a 90% confidence level. The horizontal line represents the gamma GLM central estimate and the dashed lines represent the upper and lower confidence intervals at a 90% confidence level. Central, upper and lower gamma GLM estimates do not depend on $\tilde{\theta}$.

Graph (a) in Figure 1 shows the intercept parameters $\beta_{\tilde{\theta}_0}$ and β_0 for QR and gamma GLM, respectively. The β_0 horizontal line leads to the estimation of the conditional mean of the base class 1-F, by means of Equation (A.4). The $\beta_{\tilde{\theta}_0}$ value (black points) allows the estimation of the conditional quantile of the base class compared to a selected probability level by means of Equation (3.15). Obviously, the QR intercept parameter increases as the probability level increases.

Graphs (b), (c), and (d) in Figure 1 show regression parameters for levels (2, 3, 4) of rating factor Veh_{Age}.The linear combination with the intercept parameter allows the estimation of conditional quantile and mean of the classes 2-F, 3-F, and 4-F, respectively. Focusing on risk class 2-F we have that

$$\frac{Q_{\tilde{\theta}}\left(\tilde{Y}^{(2)}\right)}{Q_{\tilde{\theta}}\left(\tilde{Y}^{(1)}\right)}=\frac{e^{(\beta_{\tilde{\theta}_0}+\beta_{\tilde{\theta}_2})}}{e^{(\beta_{\tilde{\theta}_0})}}=e^{(\beta_{\tilde{\theta}_2})}. \tag{4.3}$$

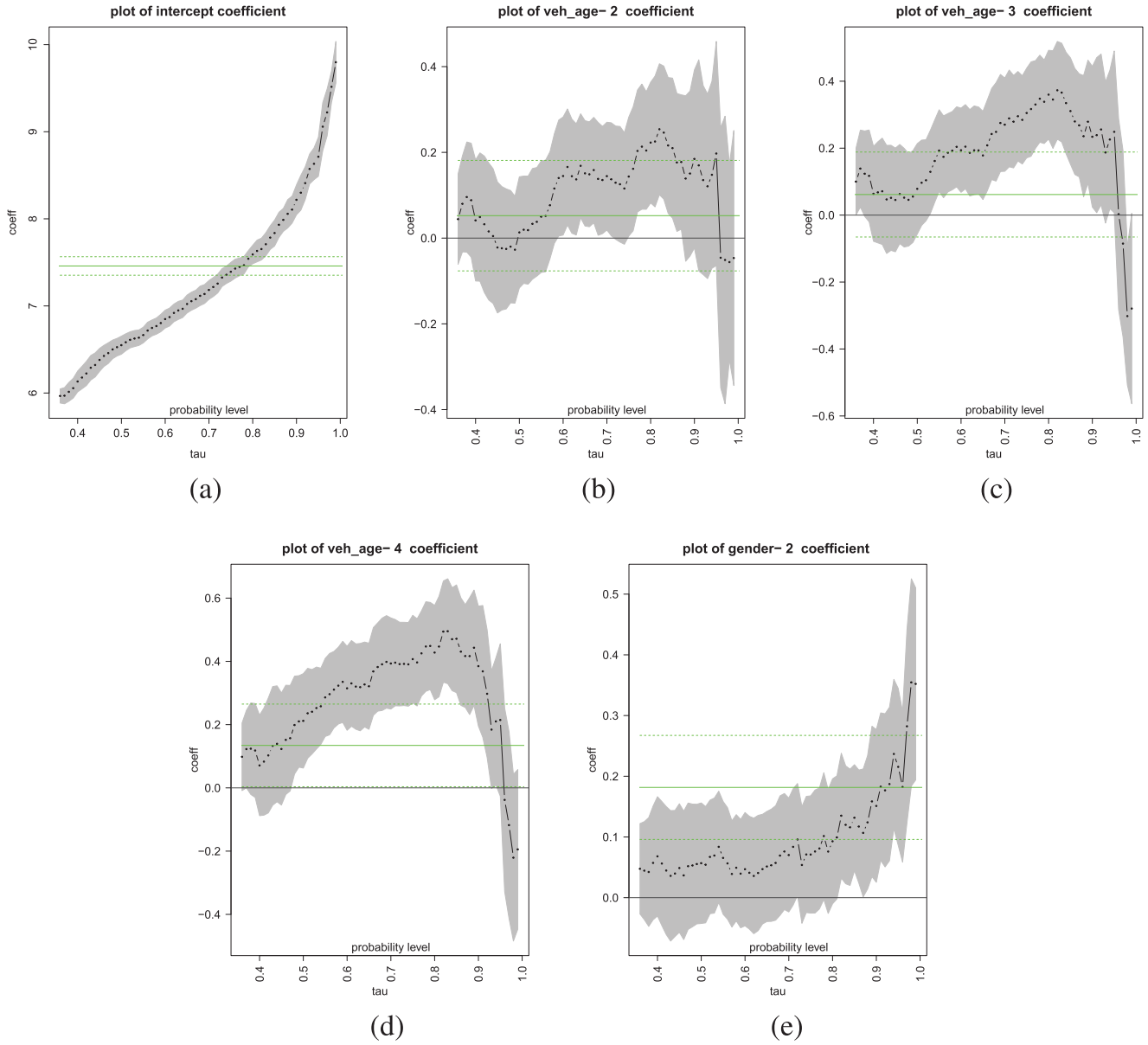


FIGURE 1. QR Parameters'Estimate : (a) Intercept; (b) $Veh_Age = 2$; (c) $Veh_Age = 3$; (d) $Veh_Age = 4$; and (e) $Gender = \mathrm{M}$.

Hence, $e^{(\beta_{\tilde{\theta}_2})}$ measures the relative difference between quantiles of the first and second risk classes. The quantile of second risk class is greater (lower) than 1 if $\beta_{\tilde{\theta}_2}$ is positive (negative). By looking at graph (b), we can observe that the regression parameter $\beta_{\tilde{\theta}_2}$ is almost positive; as a consequence, the conditional quantile of the risk class 2-F is greater than the base quantile. Moreover, parameters show an initial increase and then decrease; this means that for lower probability levels the conditional quantile is progressively greater than the base quantile. Conversely, around. 95 probability, $\beta_{\tilde{\theta}_2}$ decreases to zero and becomes negative, defining a conditional quantile closer to the base level one and smaller for extreme probability values. Finally, graph (e) represents the male gender regression parameter, $\beta_{\tilde{\theta}_4}$, which is always positive and shows an increasing trend. This means that estimated conditional quantile for males is greater than that for females, and this difference grows with probability level.

Once the regression parameters are estimated, in order to measure the estimation accuracy, Table 7 shows the following set of goodness-of-fit measures. These are frequently used measures of the differences between values predicted by a model and the values observed. The predicted values for the gamma distribution are obtained by calculating the estimator of the quantile in gamma GLM with different probability levels (i.e., 0.01, 0.02, … , 0.99).

TABLE 7
Goodness-of-Fit Measures

		RMSE $[0, +\infty)$		RSR $[0, +\infty)$		NSE $(-\infty, 1]$		φ	
h	Risk class	QR	GLM	QR	GLM	QR	GLM	Observed	GLM
1	1-F	366.61	1090.44	0.15	0.45	0.98	0.80	4.93	1.15
2	2-F	340.58	779.97	0.14	0.32	0.98	0.90	2.65	1.15
3	3-F	77.86	598.40	0.03	0.25	1.00	0.94	2.13	1.15
4	4-F	79.87	552.54	0.03	0.22	1.00	0.95	1.96	1.15
5	1-M	442.10	1677.20	0.13	0.48	0.98	0.76	4.36	1.15
6	2-M	359.20	1973.37	0.09	0.52	0.99	0.72	4.60	1.15
7	3-M	200.70	1124.74	0.06	0.36	0.99	0.87	2.71	1.15
8	4-M	290.35	1041.09	0.09	0.32	0.99	0.90	2.68	1.15

- Root mean square error (RMSE) is the square root of the mean square error (MSE). The values range from 0 to $+\infty$. If $RMSE = 0$, there is a perfect match of the modeled data to the observed data.
- Normalized root mean square error is the ratio between the RMSE and the standard deviation of the observations. It is also called RSR and ranges from 0 to $+\infty$. It is particularly useful in the comparison between datasets or models with different scales.
- Nash-Sutcliffe efficiency (NSE) is a normalized statistic that determines the relative magnitude of the residual variance compared to the measured data variance. Given a dependent variable y and its estimate $\hat{y}$, NSE can be defined as $NSE = 1 - \frac{\sum (y-\hat{y})^2}{\sum (y-E(y))^2}$.
- The NSE ranges from $-\infty$ to 1: if $NSE = 1$ there is a perfect match of the modeled data to the observed data; if $NSE = 0$ the model predictions are as accurate as the mean of the observed data; if $-\infty < NSE < 0$ the observed mean is a better predictor than the model.

As one can observe, QR provides better estimates than GLM in all of the risk classes as expected.The failure of the GLM estimates is higher as the distance between the estimated dispersion parameter $\hat{\varphi}$ and the observed dispersion parameter φ_h increases. Indeed, with reference to risk classes 3-F and 4-F, GLM reaches its best performance because of the lower discrepancy between the observed and estimated dispersion parameters. For graphical comparison that confirms the goodness-of-fit results, we show the Q–Q plot of the two models proposed in Figure 2.

These outcomes confirm the limit of the gamma GLM for risk margin estimation when the underlying distribution assumptions are violated.

4.4. Pure Premium Estimation

In order to calculate the premium rates and therefore the corresponding risk margins for each risk class, it is necessary to select the probability level $\tilde{\theta}$ for calibration of the two severity models. Following Heras, Moreno, and Vilar-Zanón (2018), it is possible to start by fixing the probability level θ (in their example equal to 99.5%) for the individual aggregate amount $Y^{(h)}$ equal for each risk class h and then, by means of (3.6), the corresponding set of $\tilde{\theta}_h$ for the severity amount $\tilde{Y}^{(h)}$. However, as stated in Section 3.3, this approach implies—in our simplified example—performing eight single QR models. I we should extend the analysis on the overall rating factors and levels contained in our dataset—as in real cases—the number of QRs to be performed should increase to 3744. These different QR models are single statistical models and each requires a specific fitting and diagnostic analysis.

For these reasons, in order to overcome over modeling and over parameterization, we suggest defining a single probability level on $\tilde{\theta}$ to model the severity amount for each $\tilde{Y}^{(h)}$ and then runing a unique QR severity model. The choice of the probability level $\tilde{\theta}$ is strategic and cannot be arbitrary.

So, as stated in 3.3, we firstly define an overall loading α with respect to the expected value. In the following, we define α in accordance with the standard formula under Solvency II for premium risk for MTPL as defined in EIOPA (2009). Therefore, we consider $\alpha = 3 \cdot \sigma \cdot E(Y)$ with $\sigma = 10\%$. The standard formula implicitly assumes measuring the distance between the VaR at a

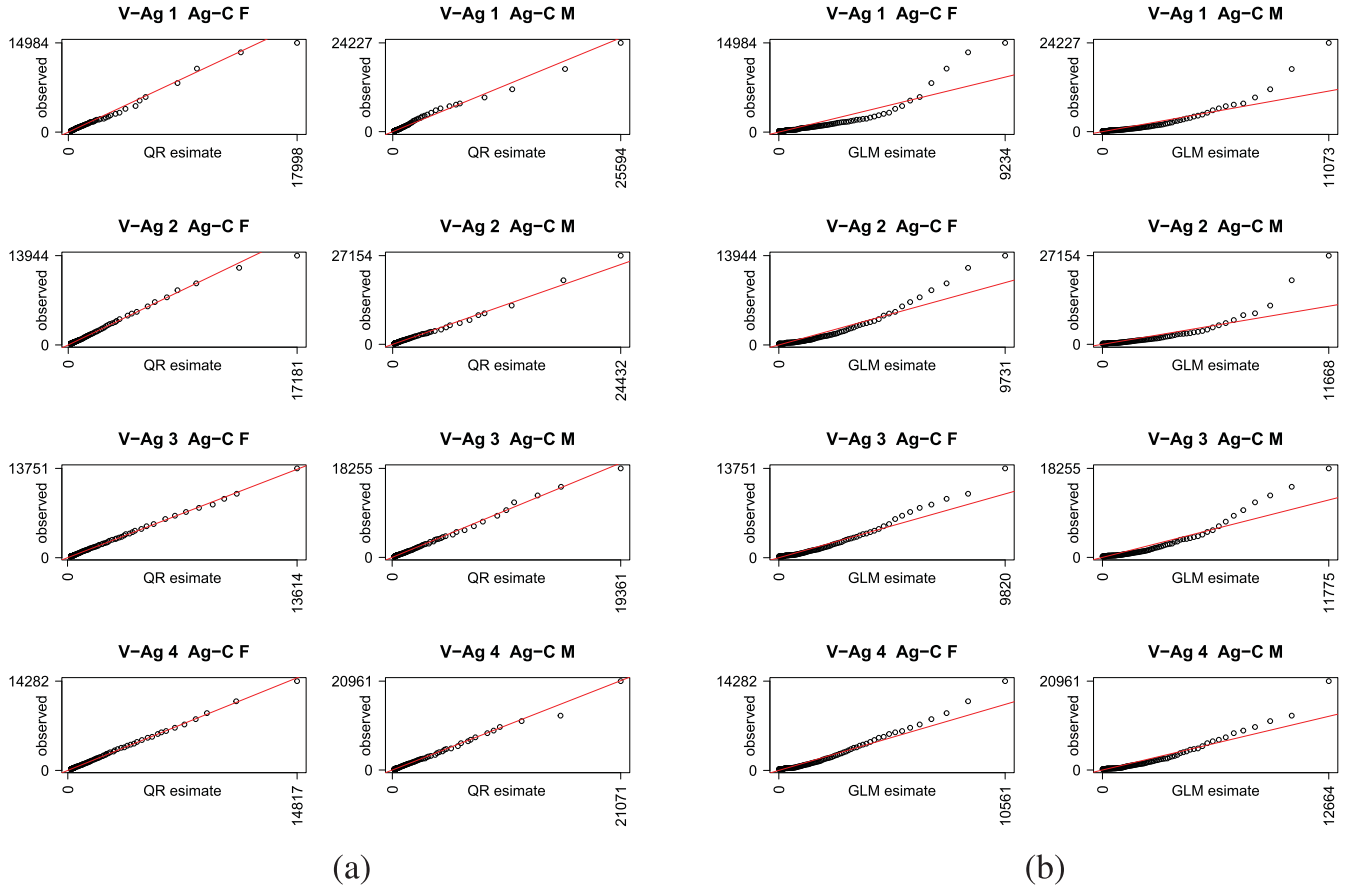


FIGURE 2. Q–Q Plot: (a) QR and (b) GLM.

99.5% confidence level and the mean of the probability distribution of aggregate claims amount by using a fixed multiplier of the volatility equal to 3 for all insurers. This assumption leads to an aggregate claims amount $\pi(Y) = 12,042,455$.

In order to define the corresponding individual probability level $\tilde{\theta}^*$ for the gamma GLM and QR, it is necessary to solve (3.17). The best solutions we find iteratively are $\tilde{\theta}^*_{GLM} = 73.2\%$ and $\tilde{\theta}^*_{QR} = 79.1\%$. Hence, we can use a single QR on the probability level $\tilde{\theta}^*_{QR} = 79.1\%$ and we can statistically analyze a unique set of parameters as shown in Table 8.

As observed, only the regression parameter $Gender = M$ is not significant.

From here on it is possible to determine the conditional quantile for the implemented multiple regression model. In Table 9 we show a comparison between the observed and estimated quantiles given the probability level $\tilde{\theta}^*_{QR}$ and $\tilde{\theta}^*_{GLM}$.

The values reported in Table 9 show that the observed and estimated quantiles are increasing as vehicle age increases and are higher for males than for females. This condition respects what has been observed in Figure 1 where the regression parameters for $Veh_Age \geq 2$ are usually higher then the base level class F-1. Moreover, because the same condition is respected for quantile estiamates via gamma GLM, it is worth noting that the percentage estimation error is very high.

Finally, pure premium ([3.2]) and risk margin are shown in Table 10.

As shown in Section 3.1, the outcomes confirm that the risk margin function obtained via gamma GLM is the same for each class and equal to the overall safety loading $\alpha = 30\%$. However, observed data exposed in (4.8) contradict this regularity of risk behavior among risk classes. QR leads to a proper distribution of the overall safety loading, defining a risk margin varying according to the riskiness of the risk class. It is worth noting that the two approaches use the same total premium income as expected due to the balance constraint in (3.17) used for the definition of the individual probability level of each model.

Nevertheless, we observe that the QR pure premium in some risk classes, such as for $h = 1$ and $h = 5$, is lower than the corresponding gamma GLM premiums (–15.50 and –22.91%), whereas for $h = 4$ it is higher $(+15.05\%)$. As intuitively one can

TABLE 8
QR Parameter Estimates

Parameter	Value	S.E	*p*Value
$\beta_{\tilde{\theta}_0}, Intercept$	7.553	0.065	
$\beta_{\tilde{\theta}_1}, Veh_{Age}=2$	0.223	0.088	.011
$\beta_{\tilde{\theta}_2}, Veh_{Age}=3$	0.343	0.082	.000
$\beta_{\tilde{\theta}_3}, Veh_{Age}=4$	0.443	0.091	.000
$\beta_{\tilde{\theta}_4}, Gender$ Male	0.072	0.064	.262

TABLE 9
Quantile Estimation QR and Gamma GLM

		$Q_{79.1\%_{QR}}(\tilde{Y}^{(h)})$			$Q_{73.2\%_{GLM}}(\tilde{Y}^{(h)})$		
h	Risk class	Observed	Estimated	Δ%	Observed	Estimated	Δ%
1	1-F	1734.21	1906.26	9.92	1413.03	2253.34	59.47
2	2-F	2395.40	2383.42	–0.50	1722.22	2374.54	37.88
3	3-F	2819.06	2686.55	–4.70	2083.06	2396.25	15.03
4	4-F	3045.05	2968.49	–2.51	2303.48	2577.13	11.88
5	1-M	2419.32	2080.32	–14.01	1760.05	2702.06	53.52
6	2-M	2505.84	2560.91	2.20	1839.54	2847.39	54.79
7	3-M	2775.10	2886.61	4.02	2063.70	2873.43	39.24
8	4-M	3074.76	3189.55	3.73	2329.05	3090.32	32.69

TABLE 10
Pure Premium and Risk Margin Estimation

			QR			GLM		
h	Risk class	Exposure	$P^{(h)}_{Q_{QR}}$	$\rho_{QR}(\tilde{Y}^{(h)})$	Total income	$P^{(h)}_{Q_{GLM}}$	$\rho{-}GLM(\tilde{Y}^{(h)})$	Total income
1	1-F	3,084.20	297.08	1.10	1,084,362	351.59	1.30	916,261
2	2-F	4,740.18	379.77	1.30	1,795,575	378.80	1.30	1,800,176
3	3-F	5,504.78	384.94	1.46	1,892,279	343.75	1.30	2,119,033
4	4-F	4,625.44	389.87	1.50	1,567,413	338.87	1.30	1,803,321
5	1-M	2,254.75	316.85	1.00	929,028	411.03	1.30	714,421
6	2-M	3,183.50	398.83	1.17	1,413,358	443.96	1.30	1,269,666
7	3-M	4,037.33	404.09	1.30	1,625,888	402.71	1.30	1,631,432
8	4-M	4,370.64	409.13	1.34	1,734,552	396.86	1.30	1,788,146
Total		31,800.82	378.68	1.30	12,042,455	378.68	1.30	12,042,455

expected, the adoption of a two-stage QR model permits performing a better risk analysis. Indeed, the riskiest risk class is $h = 8$ (4-M) and not $h = 6$ (2-M) as asserted by the equivalence premium principle or by the expected value.

5. CONCLUSIONS

This article deals with the assessment of risk margins with a quantile premium principle applied to an individual risk model. In this context, a two-step model in which the severity is measured by means of a quantile of the r.v. loss is set forth. To this aim, two multiple regression models, the GLM and the QR, are compared. The first is well known and frequently used in

actuarial sciences and statistics because it is a powerful tool for conditional mean. The latter is more robust for estimating the conditional quantile of the response variable.

GLM is considered the widely recognized industry standard actuarial method for pricing, but it is a limited tool when used to estimate the moments of the response variable and, in particular, higher quantiles. Therefore, we demonstrate that the application of a gamma GLM for the estimate of the conditional quantile of the severity amount involves an identical risk margin function; hence the quantile premium principle we analyze falls into the more traditional expected value premium principle. This is shown by an original proof in this article.

QR has been rarely discussed in the actuarial literature for ratemaking and even less in the practical field. Therefore, we contribute to the actuarial literature starting from a pure premium model that permits to spread the overall risk loading on each risk class with respect to its riskiness valuated by means of an individual security margin that respect profit or solvency constraints. With respect to Kudryavtsev (2009), the two-stage model proposed in this article permits the definition of an individual pure premium taking into account the different probability of no claim for each risk class; with respect to Heras, Moreno, and Vilar-Zanón (2018), the different risk model achieves the ratemaking goal of running a single QR. The latter is a very important feature because the goodness-of-fit and diagnostic considerations should be done only once, to avoid repeating all statistics tested by a model as many times as the number of risk classes considered in the model, which can be as high as a million in large insurance companies. Finally, we apply our approach to a real dataset comparing the accuracy of the estimation procedure between the two regression models suggested, and we confirm that two-stage QR enables a better description of the risk margin rather than a two-stage gamma GLM, as expected.

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APPENDIX A. BASIC ELEMENTS OF GLM

In this section, we will introduce basic elements of generalized linear models that are techniques that summarize many regression models that are useful in the insurance context, among others, and were introduced by Nelder and McCullagh (1989). They are widely adopted in insurance ratemaking, and for an applications readers can refer to Gigante, Picech, and Sigalotti (2010), Johansson and Ohlsson (2010), and Goldburd, Khare, and Tevet (2016). The dependent variable Y has a distribution from the exponential family with the probability density function $f(y,\kappa,\varphi) = \exp\{\frac{y\kappa - b(\kappa)}{a(\varphi)} + c(y,\varphi)\}$, where a, b, and c are known functions, and κ and φ are unknown canonical and dispersion parameters, respectively. The function $a(\varphi)$ is commonly in the form $a(\varphi) = \varphi/\omega$, where φ is constant over observation, and ω is a known prior weight that can vary from observation to observation.

The expectation and variance can be obtained under the assumption that b is twice-continuously differentiable as follows:

$$E[Y] = b'(\kappa), \quad Var[Y] = a(\varphi)b''(\kappa), \tag{A.1}$$

where the latter expression can be rewritten using the variance function, which is defined as

$$\begin{aligned} Var(\mu) &= b''[(b')^{-1}(\mu)], \\ Var(Y) &= a(\varphi)\cdot V(\mu). \end{aligned} \tag{A.2}$$

Therefore, the variance only depends on the mean value. The maximum likelihood method is used to estimate the parameters of GLMs.

Let $(\mathbf{x}_i, y_i)$ be a member of the set of observations $(i = 1, ..., r)$ where y_i is a dependent variable in the regression equation, and $\mathbf{x}_i = (x_{i1}, ..., x_{im})$ is a row vector of independent variables (covariates). The conditional mean for the i th insured is given by

$$E(Y|\mathbf{x}_i) = E(Y^{(i)}) = g^{-1}(\mathbf{x}_i \cdot \beta)\,, \tag{A.3}$$

where $\boldsymbol{\beta}$ is the column vector of regression coefficients, and the dependency of the conditional mean is described by a link function g that is strictly monotonous and twice differentiable; the choice of the link function g is bound with the features of statistical procedures.

In practical applications, the choice of the link function is often made to simplify practical considerations. In the insurance ratemaking context, a logit and/or a logarithm function is usually selected because it allows combining two models by means of a factorization of their conditional expectations. Then, for example, a log-link function Equation (A.3) could be re written as

$$E\left(Y^{(i)}\right) = e^{\mathbf{x}_i \cdot \beta}. \tag{A.4}$$

Moreover, in order to extend the analysis over to the conditional expected value of the response variable Y, it is possible to obtain an estimate of the conditional probability distribution and also estimate the dispersion parameter φ. Overdispersion is a phenomenon that is often observed in practice where the variance need not be equal to the expected value and φ scales

the relationship between the variance and the mean as in (A.2). The dispersion parameter is often unknown and its estimate could be provided in several ways. The widely recommended and apparently more robust model error estimation is based on Pearson's chi-square statistic (see Nelder and McCullagh 1989). An approximately unbiased estimator of φ is

$$\hat{\varphi}_X = \frac{\varphi \cdot X^2}{r-k} = \frac{1}{r-k}\sum_i \omega_i \frac{(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}. \tag{A.5}$$

The solution follows statistical theory where X^2 is approximately χ^2_{r-k}- distributed, with k standing for the number of regression coefficients, so that $E[X^2] \sim r-k$.

The knowledge of conditional mean and variance for each $(i = 1, ..., r)$ allows to one well-define the parameters for the exponential family distributions used in GLMs.

APPENDIX B. BASIC ELEMENTS OF QUANTILE REGRESSION

Quantile regression techniques were introduced by Koenker and Basset in 1978 and enable us to define the statistical relationship between the conditional quantile, say at level θ, and a row vector of independent covariates, $\mathbf{x}_i = (x_{i1}, ..., x_{im})$. QR looks beyond the conditional mean and allows a description of the whole conditional distribution of the response variable. According to Koenker and Bassett (1978), Koenker and Hallock (2001), and Koenker (2005), the relationship between the quantiles of the outcome and a set of explanatory variables is given by

$$Q_\theta(Y|\mathbf{x}_i) = Q_\theta\left(Y^{(i)}\right) = \mathbf{x}_i\beta_\theta, \tag{B.1}$$

where $Q_\theta(Y|\mathbf{x}_i)$ indicates the conditional quantile of random variable Y for probability θ, provided vector of covariates $\mathbf{x}_i$, and β_θ is the corresponding column vector of regression coefficients.

In statistical practice, the distribution-free approach is often used for estimation (see Koenker and Bassett 1978; Koenker and Hallock 2001; Koenker 2005; Hao and Naiman 2007). In this framework, the estimator $\hat{\beta}_\theta$ of vector β_θ from Equation (B.1) could be obtained by solving the minimization problem

$$min_{\beta_\theta \in R^m}\left\{\sum_{i:y_i \geq \mathbf{x}_i\beta_\theta} \theta|y_i - \mathbf{x}_i\beta_\theta| + \sum_{k:y_i < \mathbf{x}_i\beta_\theta} (1-\theta)|y_i - \mathbf{x}_i\beta_\theta|\right\}. \tag{B.2}$$

Several important features of the least squares regression estimator are sometimes taken for granted in elementary treatments of regression but play an important role in enabling a coherent interpretation of regression results. A basic theory of QR is described in Koenker and Bassett (1978), Koenker (2005), and Davino, Furno, and Vistocco (2014). Among these properties it is important in our context to introduce the so-called equivariance-to-monotone transformations: let $g(.)$ be a monotone nondecreasing function on $\Re$; then for any random variable Y,

$$Prob(Y \leq y) = Prob\Big(g(Y) \leq g(y)\Big). \tag{B.3}$$

The Equivariance-to-monotone transformations property is formalized, for any strictly monotonous function g, as follows:

$$Q_\theta(g(Y)) = F^{-1}_{g(Y)}(\theta) = g(F^{-1}_Y(\theta)) = g(Q_\theta(Y)). \tag{B.4}$$

Hence, the quantiles of the transformed random variable $g(Y)$ are simply the transformed quantiles of the original Y.